

Reference excerpt 01-030

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given by: $p(t + lG/c, zG) = p(t, zRA)$ (3) Where zG is the coordinate of the US scanning point and zRA is the RA coordinate. The pressure is, at the same instant tPQ , maximum in the RA ($p(tPQ, zRA)$), and equal to $p(tPQ - l/c, zRA)$ at the insonation point G. The qualitative JVP trace of 2 Figure 2: The figure a) shows the JVP trace of pressure in the IJV, measured in G, together with the ECG trace. The interval $taQP$ is represented over the curves a and QP. In figure b) it is shown the pressure along the z -axes at instant tQP . pressure in the IJV, measured in G, can be obtained simultaneously with the acquisition of the ECG trace [Sisini et al.(2016)]. On this trace, that reports both JVP and ECG waves, the time interval $taQP$ between tPQ and ta is given by: $taQP = ta - tQP = lG/c$ (4) where ta is the instant corresponding the a wave. The parameter $taQP$ is the time interval between the instant when the pressure is maximum at point G (instant ta) and the instant when the pressure is maximum at point RA (instant tQP). METHODS The applicability of the relationship expressed in Eq. 4, can be tested by comparing the instantaneous velocity ($w(t)$) of the blood in the IJV calculated according to the equation shown above, with the instantaneous blood velocity ($wD(t)$) measured with a Doppler scanner. However, the data and the